

# Limit Cycles and Self-Sustained Oscillation

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## Recall: Periodic solution and closed orbit

A nonconstant solution  $z : \mathbb{R} \rightarrow \mathbb{R}^2$  is **periodic** if there is a number  $T > 0$  such that

$$z(t + T) = z(t)$$

for every  $t \in \mathbb{R}$ . Its least positive period is its **period**. Its image

$$\Gamma = \{z(t) : 0 \leq t \leq T\}$$

is its **closed orbit**.

## Definition: Limit cycle

A closed orbit  $\Gamma$  is a **limit cycle** if it is isolated: there is a neighborhood of  $\Gamma$  containing no other closed orbit.

## Notation: Distance from an orbit

For  $z \in \mathbb{R}^2$  and a compact orbit  $\Gamma \subset \mathbb{R}^2$ , write

$$\text{dist}(z, \Gamma) = \inf_{p \in \Gamma} \|z - p\|.$$

## Definition: Orbital Lyapunov stability

A limit cycle  $\Gamma$  of  $\dot{z} = F(z)$  is **orbitally Lyapunov stable** if, for every  $\varepsilon > 0$ , there is a  $\delta > 0$  such that every solution  $z(t)$  with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every  $t \geq 0$  and satisfies

$$\text{dist}(z(t), \Gamma) < \varepsilon \quad \text{for all } t \geq 0.$$

**Definition: Attracting limit cycle**

A limit cycle  $\Gamma$  of  $\dot{z} = F(z)$  is **attracting** if there is a  $\delta > 0$  such that every solution  $z(t)$  with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every  $t \geq 0$  and satisfies

$$\text{dist}(z(t), \Gamma) \rightarrow 0 \quad (t \rightarrow \infty).$$

**Definition: Asymptotic orbital stability**

A limit cycle is **asymptotically orbitally stable** if it is both orbitally Lyapunov stable and attracting.

**Definition: Half-stable limit cycle**

A planar limit cycle  $\Gamma$  separates the plane into an inside and an outside. It is **half-stable** if it attracts every sufficiently close trajectory on one side, whereas every sufficiently close trajectory on the other side eventually leaves some fixed neighborhood of  $\Gamma$ .

**Notation: Stable limit cycle**

In these notes, **stable limit cycle** means asymptotically orbitally stable limit cycle.

**Visualization: Closed families versus isolated cycles**

| phase portrait               | isolated? | amplitude after perturbation |
|------------------------------|-----------|------------------------------|
| linear center                | no        | new amplitude persists       |
| undamped pendulum librations | no        | new energy persists          |
| attracting limit cycle       | yes       | returns to selected orbit    |

**Proposition: Radial construction principle**

Let  $I \subseteq (0, \infty)$  be an open interval, let  $R : I \rightarrow \mathbb{R}$  and  $\Omega : I \times S^1 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta), \quad (r, \theta) \in I \times S^1.$$

Suppose  $a \in I$  is an isolated zero of  $R$  and

$$\Omega(a, \theta) \neq 0$$

for every  $\theta \in S^1$ . Then  $r = a$  is a limit cycle.

**Proof.** Since  $R(a) = 0$ , uniqueness makes  $r = a$  invariant. The continuous nonzero function  $\Omega(a, \theta)$  has one sign on the connected circle, so the solution traverses  $r = a$  periodically with period

$$T = \left| \int_0^{2\pi} \frac{d\theta}{\Omega(a, \theta)} \right|.$$

On any closed orbit,  $r(t)$  is periodic. The scalar autonomous equation  $\dot{r} = R(r)$  has no nonconstant periodic solution, so  $r(t)$  must be a constant zero of  $R$ . Isolation of the zero  $a$  therefore excludes every nearby closed orbit, including noncircular ones.  $\square$

**Corollary: Radial stability test**

Let  $I \subseteq (0, \infty)$  be open, let  $R : I \rightarrow \mathbb{R}$  and  $\Omega : I \times S^1 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta).$$

Suppose  $a \in I$  is an isolated zero of  $R$  and  $\Omega(a, \theta) \neq 0$  for every  $\theta \in S^1$ . Then  $r = a$  is a limit cycle with the following classification, according to the signs of  $R(r)$  for  $r$  sufficiently close to  $a$ :

| $R(r)$ for $r < a$ | $R(r)$ for $r > a$ | type                               |
|--------------------|--------------------|------------------------------------|
| $R(r) > 0$         | $R(r) < 0$         | stable                             |
| $R(r) < 0$         | $R(r) > 0$         | not orbitally Lyapunov stable      |
| $R(r) > 0$         | $R(r) > 0$         | half-stable: attracts from inside  |
| $R(r) < 0$         | $R(r) < 0$         | half-stable: attracts from outside |

**Proof.** The sign of  $R$  determines whether nearby radii move toward or away from  $a$  on each side.  $\square$

**Question: The canonical attracting limit cycle**

Analyze

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1.$$

Interpret these as polar equations in the plane, so all coordinate pairs  $(0, \theta)$  represent the single Cartesian origin. Find and classify every nonnegative invariant radius.

**Solution.** The radial equilibria are

$$r = 0 \quad \text{and} \quad r = 1.$$

The sign chart is

| $r$       | $(0, 1)$ | $1$ | $(1, \infty)$ |
|-----------|----------|-----|---------------|
| $\dot{r}$ | +        | 0   | −             |

so the origin repels nearby nonzero states and  $r = 1$  is an attracting limit cycle. Its period is

$$T = 2\pi.$$

**Question: Solve the canonical radial equation**

For  $t \geq 0$ , solve

$$\dot{r} = r(1 - r^2), \quad r(0) = r_0 > 0.$$

**Solution.** Set  $q = r^2$ . Then

$$\dot{q} = 2q(1 - q).$$

For  $u = 1/q$ ,

$$\dot{u} = -\frac{\dot{q}}{q^2} = 2(1 - u),$$

and hence

$$u(t) = 1 + (r_0^{-2} - 1)e^{-2t}.$$

**Question: Solve the canonical radial equation (continued)**

Since  $r(t) > 0$ ,

$$r(t) = \frac{1}{\sqrt{1 + (r_0^{-2} - 1)e^{-2t}}} \rightarrow 1 \quad (t \rightarrow \infty).$$

**Question: What Cartesian system has the canonical cycle?**

Convert

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1$$

to Cartesian coordinates.

**Solution.** With

$$x = r \cos \theta, \quad y = r \sin \theta,$$

we have

$$\dot{x} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta, \quad \dot{y} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta.$$

Since  $r^2 = x^2 + y^2$ ,

$$\begin{cases} \dot{x} = x(1 - x^2 - y^2) - y, \\ \dot{y} = y(1 - x^2 - y^2) + x. \end{cases}$$

Conversely,

$$r\dot{r} = x\dot{x} + y\dot{y}, \quad r^2\dot{\theta} = x\dot{y} - y\dot{x},$$

which recover the radial and angular equations for  $r > 0$ .

**Question: Two limit cycles**

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)(4 - r^2), \quad \dot{\theta} = 1.$$

**Solution.**

|           |          |     |          |     |               |
|-----------|----------|-----|----------|-----|---------------|
| $r$       | $(0, 1)$ | $1$ | $(1, 2)$ | $2$ | $(2, \infty)$ |
| $\dot{r}$ | $-$      | $0$ | $+$      | $0$ | $-$           |

so the cycle  $r = 1$  repels nearby radii on both sides and is not orbitally Lyapunov stable, whereas  $r = 2$  is a stable limit cycle. The origin attracts every state with  $0 < r < 1$ .

**Question: A half-stable limit cycle**

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)^2(4 - r^2), \quad \dot{\theta} = 1.$$

**Solution.**

|           |        |   |        |   |                |
|-----------|--------|---|--------|---|----------------|
| $r$       | (0, 1) | 1 | (1, 2) | 2 | (2, $\infty$ ) |
| $\dot{r}$ | +      | 0 | +      | 0 | −              |

so  $r = 1$  attracts from the inside and repels from the outside: it is half-stable. The cycle  $r = 2$  remains stable.

**Theorem: Homogeneous linear systems have no limit cycles**

No planar homogeneous linear system

$$\dot{z} = Az$$

has a limit cycle.

**Proof.** Suppose  $z(t)$  is a nonzero periodic solution. For every  $c > 0$ ,

$$\frac{d}{dt}(cz(t)) = A(cz(t)),$$

so  $cz(t)$  is another periodic solution. Values of  $c \neq 1$  arbitrarily close to 1 produce distinct closed orbits arbitrarily close to the original one. Thus no closed orbit is isolated.  $\square$

**Question: What does the van der Pol energy balance prove?**

For

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \quad \mu > 0,$$

compute the derivative of the linear-oscillator energy and state its exact consequence.

**Solution.** Let

$$v = \dot{x}, \quad E(x, v) = \frac{1}{2}v^2 + \frac{1}{2}x^2.$$

Then

$$\begin{aligned} \dot{E} &= v(\dot{v} + x) \\ &= -\mu(x^2 - 1)v^2. \end{aligned}$$

Hence

$$\begin{array}{l|l} |x| < 1, v \neq 0 & \dot{E} > 0 \\ |x| > 1, v \neq 0 & \dot{E} < 0 \\ |x| = 1 \text{ or } v = 0 & \dot{E} = 0 \end{array}$$

Energy is injected exactly when  $|x| < 1$  and  $v \neq 0$ , dissipated exactly when  $|x| > 1$  and  $v \neq 0$ , and instantaneously unchanged when  $|x| = 1$  or  $v = 0$ . This pointwise sign change supplies an amplitude-selection mechanism, but the calculation alone does not prove existence, uniqueness, or stability of a periodic orbit.

**Idea: Self-sustained oscillation**

In a radial or energy-balance description of a self-sustained oscillator,

small amplitude  $\longrightarrow$  net outward motion,

large amplitude  $\longrightarrow$  net inward motion,

suggests that the dynamics may select an intermediate amplitude. This is a heuristic: it does not by itself prove that a periodic orbit exists or that such an orbit is isolated, unique, or stable. Each of those conclusions requires additional arguments.